

WHO/UNICEF JMP

Water Access Analysis

Global 2020 Report

Prepared by

JACE GLENN

DS110: Preparing Data

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1. Executive Summary

This report presents a comprehensive data analysis of global drinking water access using the WHO/UNICEF Joint Monitoring Programme (JMP) dataset for 2020. The analysis covers 210 countries and territories, examining national, urban, and rural water access across four service levels: basic, limited, unimproved, and surface water.

Key findings from the analysis:

- The dataset covers a total population of 7.778 billion people — 0.54% below the world estimate of 7.821 billion.
- 56.14% of the dataset population lives in urban areas, compared to the world estimate of 55% — a difference of 2.06%.
- Basic water access is strongly correlated with income level and urbanization across all regions.
- High-income countries average 99.55% basic national water access; low-income countries average only 62.82%.
- Rural areas consistently lag behind urban areas in basic water access across all income groups.
- The United States of America and Japan are the most populous high-income countries in the dataset.

2. Dataset Overview

2.1 Source

The dataset is sourced from the WHO/UNICEF Joint Monitoring Programme (JMP) for Water Supply, Sanitation and Hygiene, reflecting estimates for the year 2020. It includes national-level data on water service access for countries and territories globally.

2.2 Structure

The dataset contains 16 core variables across 210 rows (countries/territories):

- Identifier variables: name, income_group
- Population variables: pop_n (national population in thousands), pop_u (urban population share %)
- National water access: wat_bas_n, wat_lim_n, wat_unimp_n, wat_sur_n
- Rural water access: wat_bas_r, wat_lim_r, wat_unimp_r, wat_sur_r
- Urban water access: wat_bas_u, wat_lim_u, wat_unimp_u, wat_sur_u

2.3 Water Service Levels

The four service levels represent a hierarchy from best to worst access:

Level	Definition
Basic	Drinking water from an improved source within 30 minutes round trip collection time
Limited	Drinking water from an improved source exceeding 30 minutes collection time
Unimproved	Drinking water from an unprotected dug well or spring
Surface	Drinking water directly from a river, dam, lake, stream, or irrigation channel

3. Data Cleaning & Preparation

3.1 Import & Cleaning

The raw CSV file used a semicolon (;) delimiter for headers and comma (,) delimiter for data rows. It was imported into Google Sheets with the following cleaning steps:

- Added 16 column headers (A–P) corresponding to all variables
- Created a value_cnt column (Q) using COUNTA to validate row completeness
- Manually fixed 5 broken rows: Marshall Islands, Tonga, Cabo Verde, Sierra Leone, and Benin — all now showing value_cnt = 16
- Replaced missing values with 'NAN' for consistent handling across calculations

3.2 Derived Columns

Six additional calculated columns were added to enrich the dataset:

Column	Name	Formula / Description
R	pop_u_val	=IFERROR((pop_n * pop_u)/100, 'NAN') — Urban population value
S	pop_r	=IFERROR(100 - pop_u, 'NAN') — Rural population share %
T	pop_n (m)	=IFERROR(ROUNDUP(pop_n/1000, 0), 'NAN') — Population in millions
U	wat_bas_n (rounded)	=ROUND(wat_bas_n, 1) — Rounded to 1 decimal place
V	pop_u (rounded)	=ROUND(pop_u, 0) — Urban share rounded to whole number
W	pop_r (rounded)	=ROUND(pop_r, 0) — Rural share rounded to whole number

4. Global 2020 Population Summary

4.1 Population Metrics

Metric	Dataset Value	World Estimate	% Difference
Total Population (billions)	7.778	7.821	0.54%
Urban Population (billions)	4.367	4.302	1.50%
Urban Population Share (%)	56.14%	55%	2.06%

The dataset slightly overrepresents the urban population share compared to world estimates (56.14% vs 55%), likely reflecting the exclusion of some low-data rural territories and the NAN income group classification for small island nations.

5. Descriptive Statistics

5.1 Key Statistics for All 12 Water Features

The statistics table below summarizes Max, Min, Mean, Median, Mode, Q1, Q3, IQR, and Standard Deviation for all 12 water access features across all 210 countries:

Feature	Max	Min	Mean	Median	Q1	Q3	IQR	StdDev
wat_bas_n	100	37.2	89.86	97.34	85.64	99.88	14.24	15.09
wat_lim_n	100	0	4.81	0.54	0	4.98	4.98	11.64
wat_unimp_n	99.66	37.2	4.44	0.86	0.03	5.35	5.32	7.20
wat_sur_n	30.57	0	1.86	0	0	1.84	1.84	3.87
wat_bas_r	100	0	80.23	90.42	63.76	99.09	35.33	23.35
wat_lim_r	42.16	0	5.84	1.81	0	8.55	8.55	8.70
wat_unimp_r	51.22	0	8.74	3.23	0.16	13.27	13.11	11.84
wat_sur_r	40.52	0	4.33	0.22	0	6.16	6.16	7.21
wat_bas_u	100	0.18	93.92	98.09	92.36	99.94	7.58	11.32
wat_lim_u	100	0	4.44	0.61	0	4.28	4.28	11.74
wat_unimp_u	22.99	0	1.83	0.35	0	2.06	2.06	3.69

wat_sur_u	11.76	0	0.39	0.39	0	0.16	0.16	1.26
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5.2 Distribution Insights

For all three basic water access features (national, urban, rural), the median consistently exceeds the mean — a hallmark of left-skewed distributions:

- wat_bas_n: Mean = 89.86%, Median = 97.34% — the 7.5 percentage point gap indicates a left tail of low-performing countries pulling the average down.
- wat_bas_u: Mean = 93.92%, Median = 98.09% — urban areas perform better overall, with tighter spread.
- wat_bas_r: Mean = 80.23%, Median = 90.42% — rural areas show the widest spread (StdDev = 23.35), indicating extreme variability between countries.

Rural basic access (wat_bas_r) has the highest standard deviation (23.35) among the three basic access categories, more than double the urban equivalent (11.32). This reflects deep inequality in rural water infrastructure across countries — some rural populations have near-universal access while others remain critically underserved.

6. Income Group Analysis

6.1 Pivot Table — Average Access by Income Group

Income Group	pop_u (%)	wat_bas_n (%)	wat_lim_n (%)	wat_unimp_n (%)	wat_sur_n (%)
Low income	36.04%	62.82%	16.55%	15.21%	4.97%
Lower middle income	48.79%	82.21%	5.69%	7.90%	4.29%
Upper middle income	64.69%	96.43%	1.56%	1.48%	0.57%
High income	78.75%	99.55%	0.18%	0.25%	0.02%
Grand Total	61.53%	89.15%	4.21%	4.63%	1.99%

6.2 Trend Analysis

The pivot table reveals a consistent, monotonic relationship between income level and water access across every metric:

- Basic access increases at every income step: 62.82% → 82.21% → 96.43% → 99.55%
- Limited, unimproved, and surface access all decrease monotonically as income rises
- Urbanization (pop_u) also increases consistently: 36% → 48% → 64% → 78%

This confirms the central finding: wealth, urbanization, and basic water access are deeply correlated global indicators. Poorer, more rural countries bear a disproportionate burden of limited, unimproved, and surface water dependency.

7. Visualizations

7.1 Chart Inventory

Chart	Type	Key Insight
National Population vs Urban & Rural Share	Line Chart	Urban and rural shares fluctuate equally around 50% — mathematically guaranteed since urban + rural = 100%
Distribution of All 12 Water Features	Candlestick / Box-Whisker	Basic access features (national, urban, rural) show wide boxes at high values; limited/unimproved/surface features are flat near zero
National Water Access by Country	100% Stacked Column	Blue (basic) dominant across most countries; gaps visible in lower-income nations
Rural Water Access by Country	100% Stacked Column	More variability than national — some rural populations still rely heavily on unimproved and surface water
Urban Water Access by Country	100% Stacked Column	No consistent relationship between urban share % and access level — income is a stronger predictor
Access & Urbanization by Income Group	Bar Chart (Pivot)	Clear monotonic trend: higher income = higher basic access, lower limited/unimproved/surface access

8. Key Findings

The following findings emerge from the combined statistical and visual analysis:

1. Wealth is the strongest predictor of water access. High-income countries achieve near-universal basic access (99.55%) while low-income countries average only 62.82%.
2. Rural areas are systemically underserved. The rural basic access standard deviation (23.35) is more than double the urban equivalent (11.32), reflecting extreme inequality in rural infrastructure investment.

3. Left-skewed distributions across all basic access features confirm that most countries perform well, but a small number of severely underserved countries drag global averages down significantly.
4. Within urban areas, urbanization share alone does not predict water access levels — income group is a far stronger determinant.
5. South Sudan has the highest limited rural water access (42.16%) while maintaining above-50% urban basic access (70.01%) — illustrating the urban-rural divide within a single low-income country.
6. Countries with 0% limited rural access average 99.29% national basic access — confirming that absence of limited-tier dependency correlates with near-universal basic coverage.

9. Methodology

9.1 Tools Used

- Google Sheets — data cleaning, derived columns, descriptive statistics, charts, pivot tables
- Python (pandas, numpy) — independent validation of mean, count, and percentage calculations
- WHO/UNICEF JMP 2020 dataset — primary data source

9.2 Statistical Methods

- Descriptive statistics: Max, Min, Mean, Median, Mode, Q1, Q3, IQR, Standard Deviation computed for all 12 water features
- Percentage difference: $((\text{Dataset Value} - \text{World Estimate}) / \text{World Estimate}) \times 100$
- AVERAGEIF and COUNTIF formulas used for conditional analysis across income groups
- Pivot table aggregation: AVERAGE summarize function applied to all value fields, grouped by income_group
- NAN and regional aggregate rows excluded from income group analysis using pivot table filters

9.3 Limitations

- 47 countries have missing rural water data (NAN), with only 8 attributable to 100% urban populations — the remaining 39 have unknown data gaps
- Hong Kong SAR and Macao SAR excluded from income group analysis due to incomplete water feature data causing division errors
- Dataset urban population total (4.367B) differs from world estimate (4.302B) by 1.50%, suggesting minor coverage discrepancies
- Regional and world aggregate rows included in the raw dataset were excluded from country-level statistical calculations

10. Conclusion

The WHO/UNICEF JMP 2020 water access dataset tells one clear story: wealthier, more urbanized countries have achieved near-universal basic water access, while poorer, more rural countries still rely heavily on limited, unimproved, and surface water sources. This gap is consistent, measurable, and visible across every analytical lens applied in this report — from descriptive statistics to pivot tables to visual distributions.

The analysis demonstrates that water access inequality is not random — it follows predictable economic and geographic patterns. Addressing it requires targeted investment in low-income rural infrastructure, where both the need and the variability are greatest.

This project applied the full QNT 105 data analysis toolkit: data cleaning and preparation, derived column engineering, descriptive statistics, visualization across multiple chart types, and pivot table analysis — demonstrating proficiency in data-driven decision making from raw dataset to actionable insight.

Appendix — Google Sheets Structure

The Google Sheets workbook is organized as follows:

Sheet Tab	Contents
Raw Dataset	Original WHO/UNICEF JMP data with cleaning applied, 16 core columns + 7 derived columns (Q-W)
Global 2020 Report	Population summary table, descriptive statistics table, and all 5 charts
Income Group Analysis	Pivot table: income group vs average pop_u, wat_bas_n, wat_lim_n, wat_unimp_n, wat_sur_n + pivot chart

Dataset source: WHO/UNICEF Joint Monitoring Programme (JMP) for Water Supply, Sanitation and Hygiene — who.int/teams/unicef-who-joint-monitoring-programme-water-supply-sanitation-hygiene